

Database Modeling: Aggregation

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The Problem Aggregation Solves

Consider these facts:

- **Employee:** Aadya, Arpit
- **Project:** Database Project, AI Project
- **Relationship:** Employee ---- Works_On ---- Project

Now suppose: Manager Neha supervises: Aadya working on the Database Project.

- She is not supervising Aadya everywhere.
- She is not supervising the Database Project itself.
- She supervises: (Aadya Works_On Database Project)

That entire relationship must now participate in another relationship. This cannot be modeled using ordinary ER relationships.

What is Aggregation?

Definition: Aggregation is an abstraction in ER modeling where a relationship set is treated as a higher-level entity so that it can participate in another relationship.

Simply: Aggregation allows a relationship to behave like an entity.

Easy Analogy

Imagine a cricket tournament.

Normal relationship: Player ---- Plays ---- Match

Now suppose: An Umpire supervises Player playing a Match. The umpire isn't supervising Player alone or Match alone.

Relationship Between Relationships

This is the defining feature of Aggregation.

Normally: Entities participate in relationships.

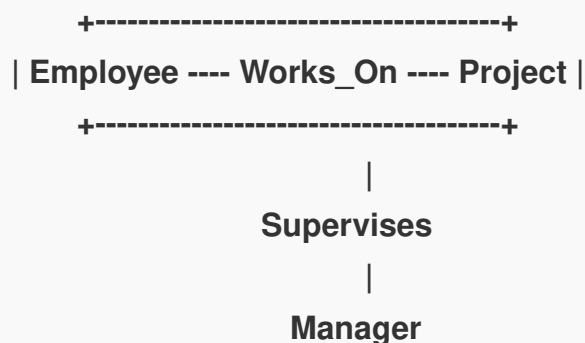
Aggregation allows: A relationship to participate in another relationship.

Example Visualization

Manager ---- Supervises ---- (Employee ---- Works_On ---- Project)

Aggregation Notation

In Chen notation, aggregation is represented by enclosing a relationship (along with the participating entities) inside a rectangle (box).



Mapping Aggregation to Relational Tables

Aggregation has no special SQL construct. Instead, it is implemented using additional tables and foreign keys.

Example Tables

Employee	Project
EmpID: 1, Name: Aadya	ProjectID: 10, Name: AI Recommendation

Works_On	Manager
EmpID: 1, ProjectID: 10, Hours: 30	ManagerID: 5, Name: Rohan

Supervises
ManagerID: 5, EmpID: 1, ProjectID: 10

Notice that Supervises references the Works_On relationship through the (EmpID, ProjectID) pair.

How do you identify when aggregation is required?

Answer: Ask this question:

"Does the new relationship involve an existing relationship rather than a single entity?"

If yes, aggregation is likely required.